Record good times

花老师班补充作业2.

1. 选(D) $0<a_n<\frac{1}{n} \Rightarrow 0<\frac{a_n}{n}<\frac{1}{n^2}$，$\sum\frac{1}{n^2}$ 收敛.

反例: (A) $\sum\frac{1}{2n}$ (B) $a_n=\begin{cases}\frac{1}{2n}, & n=2k\\ 0, & n=2k+1\end{cases} \Rightarrow \sum a_n=\sum\frac{1}{4k}$ 发散.

(C) $\sum\frac{1}{\sqrt{2n}}$.

2. 选(C).

$\sum u_n$，$\sum v_n$ 发散 $\Rightarrow \sum|u_n|$，$\sum|v_n|$ 发散. $\Rightarrow \sum(|u_n|+|v_n|)$ 发散.

反例: (A) $\sum\left[\frac{1}{n}+(-\frac{1}{n})\right]=\sum 0$ 收敛.

(B) $\sum\frac{1}{n}\cdot\frac{1}{n}=\sum\frac{1}{n^2}$ (D) $\sum\left(\frac{1}{n^2}+\frac{1}{n^2}\right)=\sum\frac{2}{n^2}$

3. 选(D). $S_n=u_1+u_2+\cdots+u_n \to u$. since $\sum u_n$ 收敛. and $u_n\to 0$.

$$\begin{aligned}\Rightarrow \text{let } A_n=\sum_{k=1}^{n}(u_k+u_{k+1}) &=(u_1+u_2)+(u_2+u_3)+\cdots+(u_n+u_{n+1})\\ &=(u_1+\cdots+u_n)+(u_2+\cdots+u_{n+1})\\ &=S_n+S_{n+1}-u_1\\ &\to u+u-u_1=2u-u_1.\end{aligned}$$

$\Rightarrow \sum(u_n+u_{n+1})$ 收敛.

反例: (A) $u_n=(-1)^n\frac{1}{\ln n}$，$\sum u_n$ 条件收敛 $\Rightarrow \sum(-1)^n\frac{u_n}{n}=\sum\frac{1}{n\ln n}$ 发散.

(B) $u_n=(-1)^n\frac{1}{\sqrt{n}}$，$\sum u_n$ 条件收敛 $\Rightarrow \sum u_n^2=\sum\frac{1}{n}$ 发散

(C) $u_n=(-1)^{n+1}\frac{1}{n}$，$\sum(u_{2n-1}-u_{2n})$，$S_n=(u_1-u_2)+(u_3-u_4)+\cdots+(u_{2n-1}-u_{2n})$
发散.
$=1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\cdots+\frac{1}{2n-1}+\frac{1}{2n}$
$=\sum_{k=1}^{2n}\frac{1}{k}\to\infty$

4. 选(C).

$\left|\frac{\sin a}{n^2}\right| \le \frac{1}{n^2}$, $\sum\frac{1}{n^2}$ 收敛 $\Rightarrow$ 绝对收敛., $\sum\frac{1}{\sqrt{n}}$ 发散 $\Rightarrow$ 发散.

8. 选(A).

$I=\sum(-1)^n\ln(1+\frac{1}{n})$ 条件 $\begin{cases}\frac{\ln(1+\frac{1}{n})}{\frac{1}{n}}\to 1,\ \sum\ln(1+\frac{1}{n})\text{ 发} \\ \ln(1+\frac{1}{n})>0,\ \to 0,\ \downarrow\end{cases}$

$J=\sum\ln^2(1+\frac{1}{n})$ 收敛 $\frac{\ln^2(1+\frac{1}{n})}{\frac{1}{n^2}}\to 1$,

34. (3) $\ln n < n^{\frac{1}{6}} \Rightarrow \frac{(\ln n)^2}{n^{\frac{3}{2}}} < \frac{n^{\frac{1}{3}}}{n^{\frac{3}{2}}} = \frac{1}{n^{\frac{3}{2}-\frac{1}{3}}} = \frac{1}{n^{\frac{7}{6}}}$

$\Rightarrow$ 绝对收敛.

(4) $\left|2^n\sin\frac{\pi}{3^n}\right| \le \left(\frac{2}{3}\right)^n\pi$, $\Rightarrow$ 绝对收敛.

(5) $\sqrt[n]{a_n} = \frac{n!}{n^2} = \frac{(n-1)!}{n} = \frac{n-1}{n}\cdot(n-2)! \to \infty > 1$

$\Rightarrow$ 发散.

(6) $\lim\limits_{n\to\infty}\frac{\frac{1}{n(\ln(1+\frac{1}{n})+1)}}{\frac{1}{n}} = \lim\limits_{n\to\infty}(\ln(1+\frac{1}{n})+1)\to 1$

$\Rightarrow$ 发散.

(7) $\lim\limits_{n\to\infty}\frac{1-\cos\frac{\lambda\pi}{n}}{\frac{1}{2}\cdot\frac{\lambda^2\pi^2}{n^2}} = 1$, $\sum\frac{1}{2}\cdot\frac{\lambda^2\pi^2}{n^2} = \frac{\lambda^2\pi^2}{2}\sum\frac{1}{n^2}$ 收敛

$\Rightarrow$ 收敛.

59. (2) $\lim_{n\to\infty}\frac{a_{n+1}}{a_n} = e\cdot\left(\frac{n}{n+1}\right)^n = \frac{e}{(1+\frac{1}{n})^n} \to 1.$ ✗

Since $\frac{a_{n+1}}{a_n} = \frac{e}{(1+\frac{1}{n})^n}$, $(1+\frac{1}{n})^n \to e$, $(1+\frac{1}{n})^n \uparrow \Rightarrow (1+\frac{1}{n})^n < e.$

$\Rightarrow \frac{a_{n+1}}{a_n} > 1 \Rightarrow a_n > a_1 = e \quad \lim_{n\to\infty} a_n \geq e$

$\Rightarrow \lim_{n\to\infty} a_n \neq 0.$

(4) $\sum (-1)^{n+1}\frac{\ln n}{n^p}$

if $0<p\leq 1 \Rightarrow \frac{\ln n}{n^p} > \frac{1}{n^p}$, $\sum\frac{1}{n^p}$ diverges since $0<p<1$

$\Rightarrow \sum\frac{\ln n}{n^p}$ diverges. $+\begin{cases}\frac{\ln n}{n^p}>0, \to 0 \downarrow\end{cases} \Rightarrow$ 条件收敛.

if $p>1 \Rightarrow \ln n < n^c \Rightarrow \frac{\ln n}{n^p} < \frac{n^c}{n^p} = \frac{1}{n^{p-c}}$, $p-c>1$ $\Rightarrow 0<c<p-1$

let $c=\frac{p-1}{2} \Rightarrow \frac{\ln n}{n^p} < \frac{1}{n^{\frac{p+1}{2}}}$, $\frac{p+1}{2}>1$

$\Rightarrow \sum\frac{\ln n}{n^p}$ 收敛 $\Rightarrow$ 绝对收敛.

(6) $\left|(-1)^n 2^n \sin\frac{\pi}{3^n}\right| \leq \frac{2^n}{3^n}\pi = \pi\cdot\left(\frac{2}{3}\right)^n$, $\pi\sum\left(\frac{2}{3}\right)^n$ 收敛 $\Rightarrow$ 绝对收敛.

(8) $\lim_{n\to\infty}\frac{n^{n+\frac{1}{n}}}{(n+\frac{1}{n})^n} = \lim_{n\to\infty}\frac{\sqrt[n]{n}}{(1+\frac{1}{n^2})^n} = \lim_{n\to\infty}\frac{1}{\left[\underbrace{(1+\frac{1}{n^2})^{n^2}}_{\to e}\right]^{\frac{1}{n}}} = 1 \neq 0$

$\Rightarrow$ 发散.